



HWA CHONG INSTITUTION
2019 C1 H2 CHEMISTRY PROMOTIONAL EXAMINATION
SUGGESTED SOLUTIONS

Paper 1

1	2	3	4	5	6	7	8	9	10
C	D	D	A	C	A	C	A	B	A
11	12	13	14	15	16	17	18	19	20
B	C	D	B	A	B	D	C	B	D

Comments

1	C	This question tests on the understanding of electronic configurations
	A	[He]2s ² 2p ⁵ – 1 unpaired electron in 2p subshell
	B	[Ar]3d ¹⁰ 4s ² – 0 unpaired electrons
	C	[Ar]3d ⁷ – 3 unpaired electrons in the 3d subshell
	D	[Ne]3s ² 3p ⁴ – 2 unpaired electrons in 3p subshell

- 2 D The use of the Data Booklet for this question is optional but one would need to refer to the Periodic Table to compare the elements and their trends in atomic radii, ionic radii and first ionisation energies.

All X and Y pairs are from the same period. Since X is a non-metallic element and Y is a metallic element, the ionic radii for X will naturally be bigger than that of Y.

Atomic radius decreases across the period. This applies to all 3 options

The first I.E. of Z being higher than that of Y is relatively ok for options 1 and 3. Option 2 is tricky as the 1st I.E. of Mg is higher than that of Al since the outermost electron in Al is in the further 3p subshell.

- 3 D The N atoms which have two N–C single bonds each possesses two lone pairs (to fulfil octet), and hence exist as N[–] ions. The two Mg–N bonds drawn using solid lines are hence ionic bonds.

On the other hand, the N atom which form a N=C double bond and a N–C single bond would only have one lone pair, and is hence electrically neutral. This N atom can use its lone pair for dative bond formation with Mg. The two Mg–N bonds drawn using dotted lines are hence dative covalent bonds.

A	With a planar arrangement of N atoms around Mg, Mg cannot be sp ³ hybridised (this would have given a tetrahedral geometry). Thus option A is incorrect.
B	There is no side-ways overlap of p-orbitals around the Mg atom, so there is no π bonding. Option B is incorrect.
C	Metallic bonding only exists in metals, and thus is not applicable to the bonding between Mg and N. Option C is incorrect.
D	The two Mg–N bonds drawn using dotted lines are dative covalent bonds, as explained above.

- 4 A Let final pressure = p₂

$$\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2}$$

$$\frac{2.05 \times 10^4 \times 50}{220+273} = \frac{p_2 \times 150}{18+273}$$

$$p_2 = 4.03 \times 10^3 \text{ Pa}$$

- 5 C No of moles of FeC₂O₄ = (20/1000)(0.020) = 0.00040 mol
 No of moles of KMnO₄ = (15/1000)(0.020) = 0.00030 mol

Note that both Fe²⁺ and C₂O₄^{2–} are both oxidised by KMnO₄
 1 mol Fe²⁺ gives out 1 mol e and 1 mol C₂O₄^{2–} gives out 2 mol e

Total amount of electrons transferred = 0.00040 x 3 = 0.0012 mol
 Amount of electrons taken in by 1 mol KMnO₄ = 0.0012 / 0.00030 = 4

Initial oxidation state of Mn in KMnO₄: +7
 Final oxidation state of Mn after taking in 4 mol of e: (+7) – 4 = +3

- 6 A The NaOH added is used to determine the amount of H⁺ ions present in the anion of X.

$$\text{No of moles of H}^+ = (24/1000)(0.100) = 0.00240 \text{ mol}$$

The AgNO₃ added is used to determine the amount of Cl[–] in the anion of X.

$$\text{No of moles of Cl}^- = (48/1000)(0.100) = 0.00480 \text{ mol}$$

$$\text{Ratio of H}^+ \text{ to Cl}^- = 0.0024 : 0.0048 = 1 : 2$$

Hence the formula for anion X is HC₂O₄^{2–}.

- 7 C KOH and HNO₃ reacts in a 1:1 mole ratio. 0.0025 mol of H₂O is produced in this neutralization reaction. Heat evolved is directly proportional to the number

of moles of water produced since volume for both experiments is constant at 75 cm³.

0.0025 mol of H₂O produced in 75 cm³ reaction mixture increases the temperature by 8.7 °C.

H₂SO₄ is the limiting reagent here and 0.0025 mol of H₂O will be produced in 75 cm³ reaction mixture. Thus temperature rise will be the same at 8.7°C

8 A $\Delta G = \Delta H - T\Delta S$

For a reaction to be spontaneous at all temperatures, $\Delta H < 0$ and $\Delta S > 0$.

A	$\Delta S > 0$ since there are more moles of gas after reaction. $\Delta H < 0$ $\Delta G < 0$ at all temperatures.
B	$\Delta S < 0$ since there are lesser moles of gas after reaction. $\Delta H < 0$ Thus, $\Delta G < 0$ only at low temperatures.
C	$\Delta S > 0$ since there are more moles of gas after reaction. $\Delta H > 0$ $\Delta G < 0$ only at high temperatures.
D	$\Delta S > 0$ since there are more moles of gas after reaction. $\Delta H > 0$ $\Delta G < 0$ only at high temperatures.

9 B 1 is incorrect as $\Delta H_f^\circ(\text{CO}_2)$ is represented by the equation $\text{C(s)} + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g})$.

2 is incorrect as $\Delta H_r = \sum m\Delta H_c(\text{reactants}) - \sum n\Delta H_c(\text{products})$. Hence, you need to multiply the enthalpy change of combustion of hydrogen by 2.

3 is correct. $\Delta H_r = \sum m\Delta H_f^\circ(\text{products}) - \sum n\Delta H_f^\circ(\text{reactants})$. ΔH_f° of sulfur and oxygen are zero since they are elements in their standard states.

10 A	A With more surface area, there are more reactants exposed to collisions, increasing the frequency of collisions and hence chance of effective collisions.
	B There is no change in activation energy or temperature of the reaction, hence the rate constant remains the same.
	C There is no change in the mechanism of the reaction or catalyst added. The activation energy remains the same.

D	There is no change in temperature of the system. The amount of energy and energy distribution in the system remains unchanged.
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11 B Catalytic convertor removes unburnt hydrocarbons, carbon, carbon monoxide, and oxides of nitrogen in the exhaust.

12 C The total area under the Boltzmann distribution curve, which represents the total number of molecules, does not change when temperature changes. Hence, P+Q represents the total number of molecules at T₁ as well as at T₂.

The number of molecules to the right of the vertical line representing activation energy has energy greater than or equal to the activation energy of a reaction. At T₂, this is represented as Q+R.

Hence the proportion of molecules with energy greater than or equal to activation energy is $\frac{Q+R}{P+Q}$.

13 D The equilibrium constant K is only affected by temperature, hence options A, B and C are all wrong. Since the forward reaction is exothermic, reducing the temperature will cause the position of equilibrium to shift to the right to produce more heat. This will thus lead to an increase in the production of ammonia.

14 B Statement 1 is an incorrect statement. When NaCl is added, the Cl⁻ ions will form AgCl precipitate with the Ag⁺ ions. Hence the concentration of Ag⁺ falls, and causes the position of equilibrium to shift to the left.

Statement 2 is incorrect. The K_c has units of mol⁻¹ dm³.

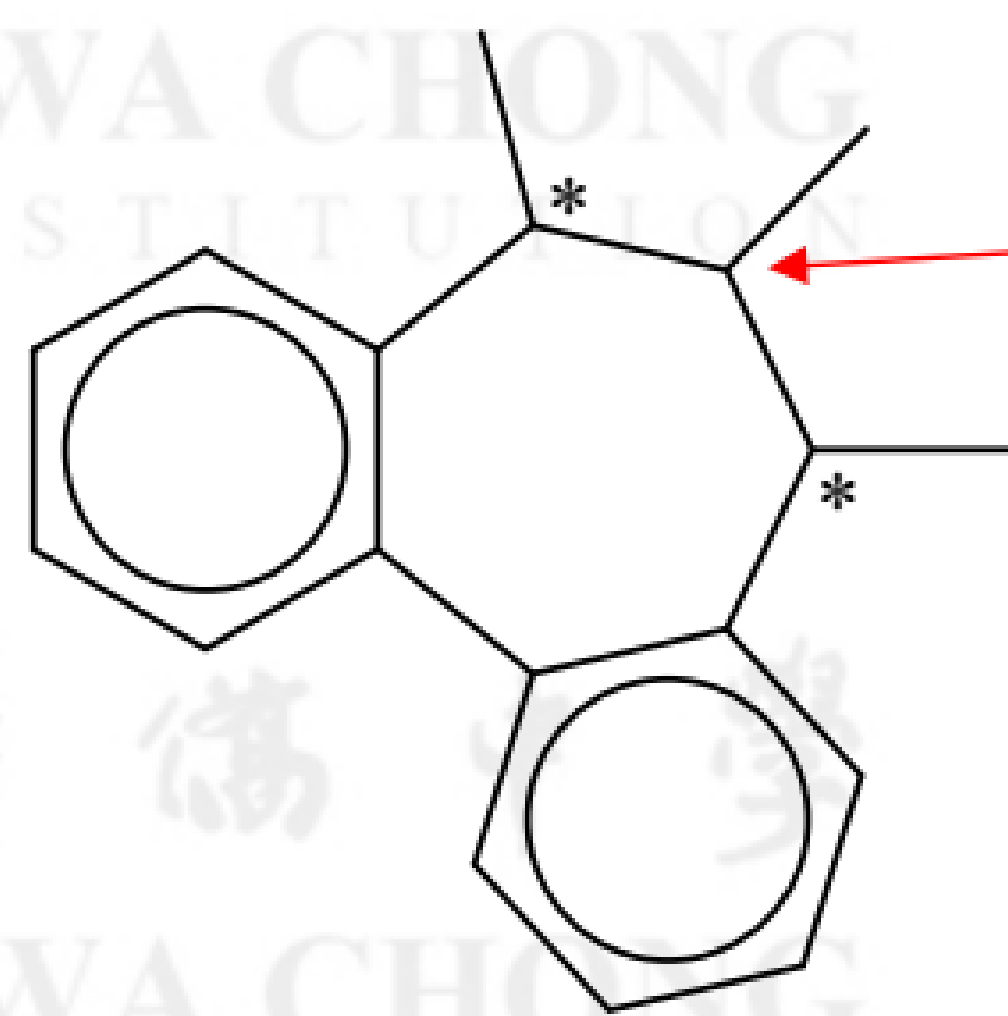
Statement 3 is a correct statement. The K_c value drops from 3.1 to 1.2 when the temperature is increased. This implies that the increase in temperature has favoured the backward reaction which is endothermic. The forward reaction is thus exothermic.

15 A Condensation reactions are reactions in which two molecules combine to form a larger molecule, producing a small molecule such as H₂O or HCl as a by-product. Options B and C are hence examples of condensation reactions.

Option A is an elimination reaction.

Option D is a condensation polymerisation reaction.

16 B



Not a chiral centre as there is a plane of symmetry through this carbon.

A chiral centre is an atom with 4 different groups bonded to it. The chiral centres are marked with * above.

17 D Students should appreciate that free radical reactions are uncontrollable in nature, and it is not possible to restrict the possible reactions that can happen in the system.

Chloromethane, CH_3Cl , that is formed can undergo further substitution, and the following radicals can be formed in the process:

- CH_2Cl (formed by breaking a C-H bond in CH_3Cl)
- CHCl_2 (formed by breaking a C-H bond in CH_2Cl_2)

The combination of • CHCl_2 and • CH_3 gives option B, while combination of two • CHCl_2 gives option C.

Hence, all four products are possible.

18 C When ethene is reacted with bromine water, Br_2 acts as the electrophile and attacks the $\text{C}=\text{C}$ double bond. This leads to the formation of the carbocation $[\text{CH}_2\text{CH}_2\text{Br}]^+$. As such, all the products formed would have a Br atom attached to one of the carbons.

Option 1 cannot be formed as there is no Cl_2 electrophile present.

Option 2 can be formed when Cl^- attacks the $[\text{CH}_2\text{CH}_2\text{Br}]^+$ carbocation.

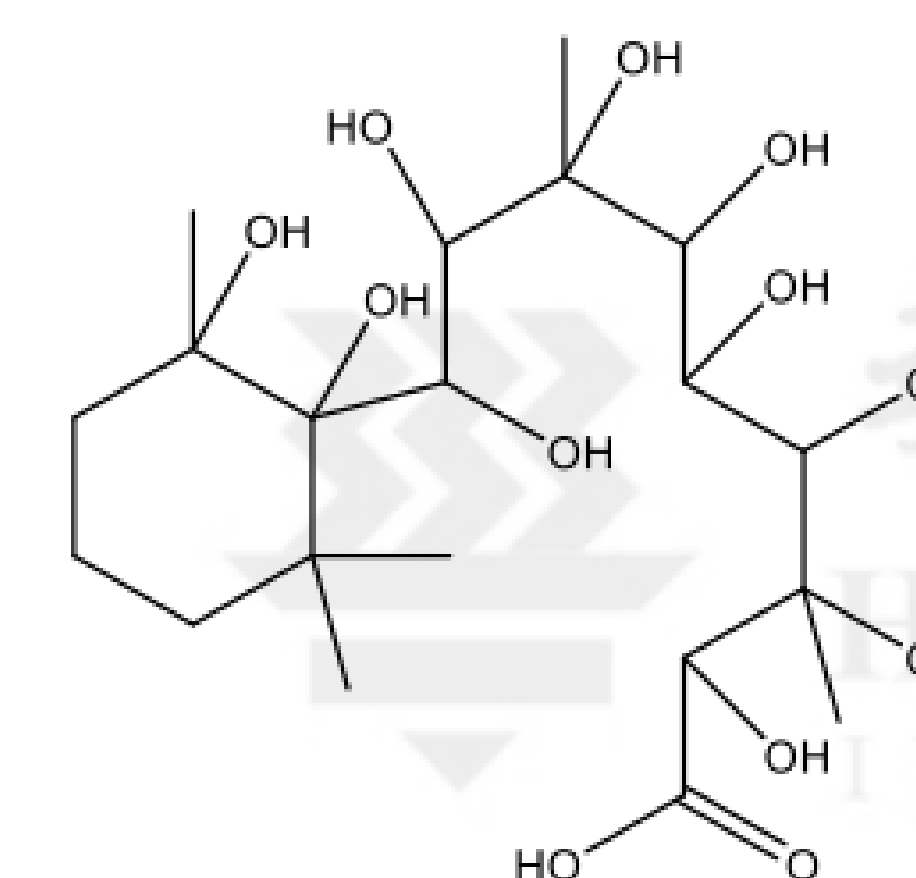
Option 3 can be formed when H_2O attacks the $[\text{CH}_2\text{CH}_2\text{Br}]^+$ carbocation.

19 B

A	<i>Alitretinoin</i> does not have any chiral carbons but there can be cis-trans about 4 $\text{C}=\text{C}$ double bonds. Hence, it can have $2^4 = 16$ cis-trans isomers.
B	In the presence of excess bromine and UV light, the alkyl chains can undergo free radical substitution. However, the alkenes within the molecule can also undergo an electrophilic addition with the bromine present. Hence this statement is the incorrect one.

C

When cold dilute potassium manganate(VII) is added, the alkenes in *Alitretinoin* undergo mild oxidation to form diols. The structure of the product is as shown and it contains chiral carbons.



D

Addition of hot KMnO_4 will lead to oxidative cleavage of the alkenes. The ethanoic acid produced will be further oxidized to form CO_2 .

20 D

A

The nitro group after the Step 1 will direct the chlorination to the 3-position which is desired. However, the alkylation in Step 3 will also be directed by the nitro group to the 3-position, despite conflicting directing effects from the chlorine to the 2,4-position. This conflict in directing effects will lead to a lower yield of the desired product.

B

If nitration is the first step, the nitro group will direct the alkylation to the 3-position on the benzene ring instead of the 4-position. Hence yield of the desired product will be lowered.

C

The methyl group after Step 1 will direct the chlorination to the 2,4 position. Subsequent Step 3 can either be on the 3 or 4-positions due to conflicting directing effects from the methyl and chlorine substituent groups.

D

The methyl group after Step 1 will direct the nitro group to the 2,4-positions, 4th position being favoured due to steric effect. In step 3, both the nitro group and the methyl group will direct the chlorination at the 2-position. There is no conflicting directing effects for this synthetic route.

Paper 2

Mr Wee CS: S60 – S6F; S78 – S7A

Mr Li JX: S6G – S6Q; S70 – S77; S7B – S7J

- 1 (a) Ideal gases have negligible intermolecular forces whereas real gases have significant intermolecular forces. [1]

Ideal gas particles have negligible volume compared to the volume of the container, while real gas particles have significant volume compared to the volume of the container. [1]

(Reject: All collisions of ideal gas particles are perfectly elastic.)

Students are expected to remember the assumptions for ideal gases. When asked for two assumptions, students should write the 2 more important ones related to intermolecular forces and volume of the gas particles.

Only when three assumptions are asked for by the question is the perfectly elastic collision used as an answer (for the third assumption).

An alarmingly large number of students are still unable to distinguish between volume of the gas (which refers to the volume of the container occupied by the gas), and volume of the gas particles. For an ideal gas, it is the particles that have negligible volume compared to the volume occupied by the gas; so DON'T write statements like '*the gas has negligible volume*'. If the gas had negligible volume, then there would not be a need to study it.

- 1 (b) (i) Assume 1 mol NH₃ is present initially.

	2NH ₃ (g)	=	N ₂ (g)	+	3H ₂ (g)
I/mol	1		0		0
C/mol	-0.15		+0.075		+0.225
E/mol	0.85		0.075		0.225

$$\text{Partial pressure of ammonia} = \frac{10}{0.075} \times 0.85 = 113 \text{ kPa [1]}$$

OR

Let initial partial pressure of NH₃ be x kPa.

	2NH ₃ (g)	=	N ₂ (g)	+	3H ₂ (g)
I/kPa	x		0		0
C/ kPa	-0.15x		+0.075x		+0.225x
E/ kPa	0.85x		0.075x		0.225x
			= 10		

$$x = 10 / 0.075 = 133.33$$

$$\text{Partial pressure of ammonia} = 0.85x = 113 \text{ kPa}$$

The use of ICE table is not necessary but is helpful in keeping track of the dissociation of the ammonia gas.

- 1 (b) (ii) Partial pressure of hydrogen = $\frac{10}{0.075} \times 0.225 = 30 \text{ kPa}$

$$\text{Total pressure} = 113 + 10 + 30 = 153 \text{ kPa [1] ecf from (b)(i)}$$

The total pressure of the system is the sum of the partial pressure of each individual gas that makes up the system. From the partial pressure of N₂, the partial pressure of H₂ can be calculated (it's three times the partial pressure of N₂).

Alternative methods may be used to arrive at the answer.

- 1 (b) (iii) The ammonia molecules adsorb onto the surface of the iridium metal. [1]

The (intramolecular) covalent bonds within the ammonia molecules are weakened [0.5], reducing the activation energy needed to break the bonds. [0.5]

Ammonia is broken down and nitrogen and hydrogen molecules are formed. Nitrogen and hydrogen molecules desorb from the surface of the iridium. [1]

Students are expected to be able to describe the mode of action for a heterogeneous catalyst and understand the processes that happen during the catalysis. This is a specific learning objective in the syllabus.

Many students were unable to demonstrate that they understand it is the covalent bonds within the molecule that are weakened, that leads to a decrease in activation energy. Many **wrongly** thought it was the intermolecular forces/ hydrogen bonding that are weakened that leads to the lowering of activation energy.

Good quality answers were specific enough to identify that it was the N–H bonds in ammonia that were weakened.

Some students did not understand that adsorption is a surface process and **wrongly** thought that it involves absorbing reactants into the material of the catalyst. The catalysis only occurs on the surface of the catalyst. Either that, or students did not appreciate that a difference in spelling (adsorb vs absorb) means very different processes.

Students are often confused between the spelling of the verb (adsorb, desorb) and the noun (adsorption, desorption).

Some students confused heterogeneous catalyst with an enzyme, which acts in a completely different manner, even though there are some similarities in their behavior, like saturation of active sites.

1 (c) $\Delta G = \Delta H - T\Delta S$ [0.5]

Entropy change would be positive since there is an increase in the number of gaseous molecules. [0.5]

At higher temperature, $|T\Delta S|$ would be even greater than $|\Delta H|$, so ΔG becomes more negative. [0.5]

Hence, the decomposition is still/remains spontaneous. [0.5]

The discussion of temperature dependence of spontaneity based on enthalpy and entropy changes is a specific learning objective in the syllabus. When presented with this type of question, you should immediately realise you need to apply the equation that relates these variables and cite the equation.

Students should then look for clues within the question that may provide information on the enthalpy and entropy changes. Students should then discuss the effect in the context of the question.

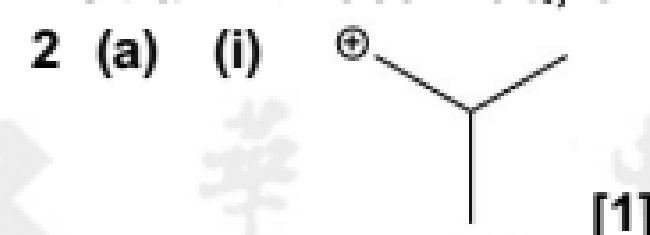
In this question, the scenario is: the process is already spontaneous, meaning the ΔG is already a negative value. The enthalpy change is given directly in the question as positive, hence an endothermic process. The entropy change is indirectly given in the chemical equation, with more gases produced at the end of the reaction; hence students need to mention why the entropy change is positive.

Your answer should then make use of ΔH and $T\Delta S$ to discuss the impact of the change in T on ΔG . In this context, the ΔG became more negative since it was already negative to begin with.

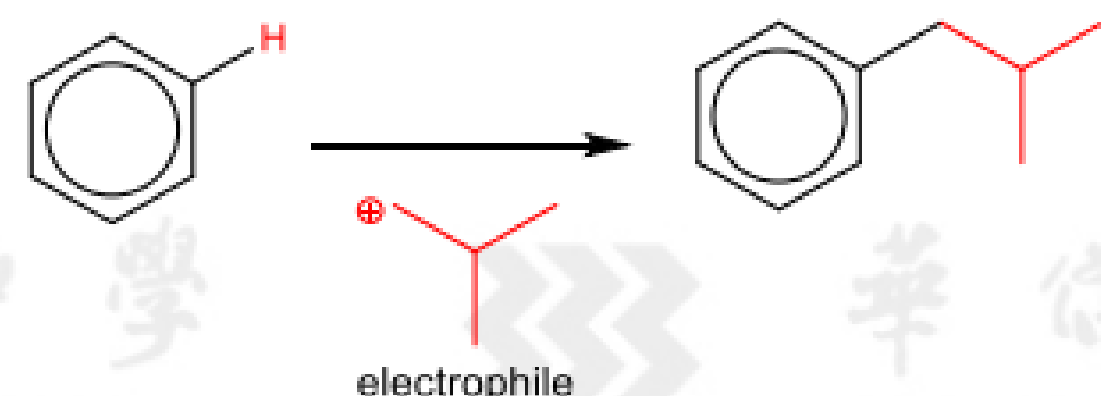
Hence, the reaction remains spontaneous with increase in temperature.

Ms P Ho: S60 – S6H; S70 – S75; S7F – S7J

Ms Fan YT: S6J – 6Q; S76 – S7E

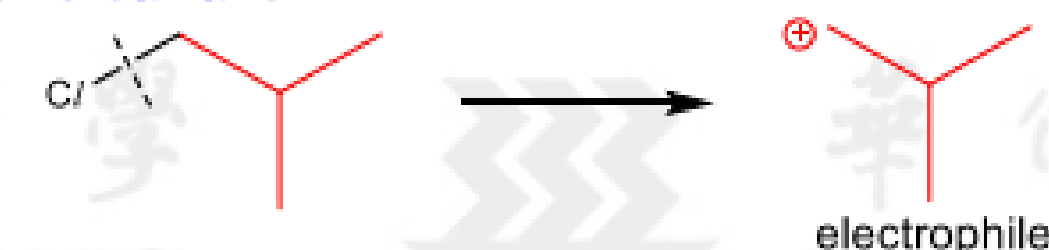


The electrophile to be substituted onto the benzene can be worked out by looking at the final substituted product and working backwards, as shown below.

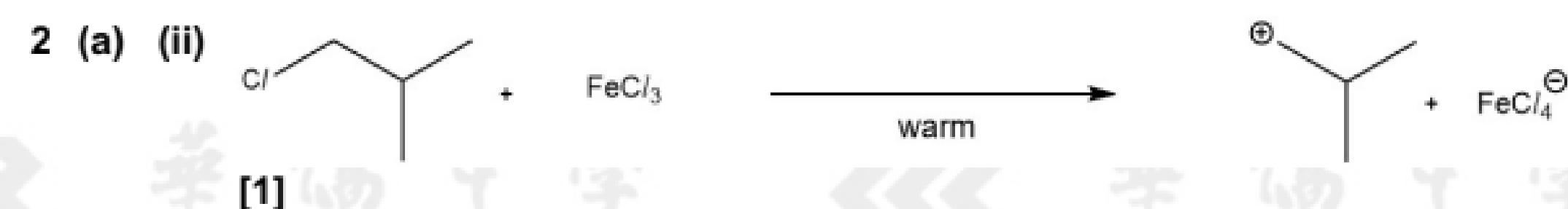


The positive charge must be written next to the correct carbon instead of an overall + charge. The correct carbocation must be given before the full mark is awarded.

The electrophile can also be derived from the equation, by removing a chloride anion from the chloroalkane.



Do take note when counting the number of carbon atoms; many students added an extra carbon by mistake as they were not used to working with skeletal structures. Full structural formula was also accepted.



This is an application question. Students are to apply their knowledge of the generation of electrophile in the standard Friedel-Crafts Alkylation, or chlorination, of benzene, given in the Lecture Notes. Since FeCl_3 acts as a Lewis acid catalyst, FeCl_4^- will be produced in the first step to generate the strong electrophile needed to react with the benzene.

ECF is awarded for this question provided the equation is correctly balanced and the electrophile drawn is the same as the incorrect electrophile in part (a)(i).

2 (a) (iii) Fe in FeCl_3 is electron deficient, hence it can accept a lone pair of electrons from Cl^- . [1]

This is based on the definition of a Lewis acid. Students are required to state clearly that a pair of electrons or 2 electrons are accepted.

2 (a) (iv) Dilute KMnO_4 , aq H_2SO_4 , heat [1]

Compound A will decolorise purple KMnO_4 while for compound B, KMnO_4 remains purple. [1]

Compounds A and B are both alkylbenzenes, however, B contains a tert-butyl alkyl group with no H atom on the C adjacent to benzene. Hence, B cannot undergo side-chain oxidation while A can.

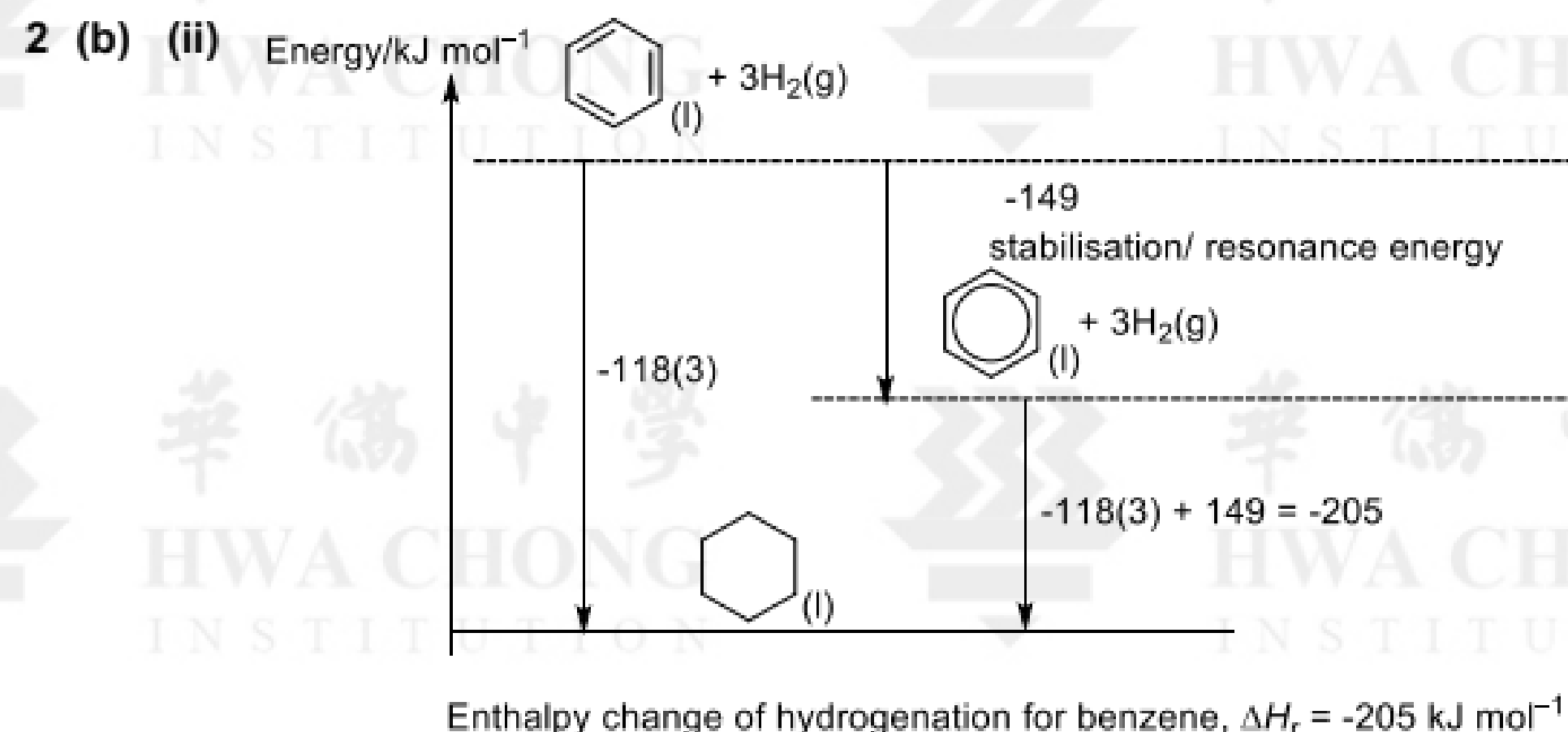
Students are expected to know the reagents and conditions for side-chain oxidation of alkylbenzene. Dilute or aqueous has to be stated for sulfuric acid. Some students used alkaline conditions, "Dilute KMnO_4 , aq NaOH, heat/warm", which was accepted. However, for observations using alkaline conditions, students are expected to state that the purple solution decolourises and a brown precipitate of MnO_2 is formed. No marks were awarded if student merely stated decolourisation of purple solution when alkaline conditions were used.

The initial "purple" colour has to be clearly stated. Merely saying KMnO_4 decolourises without any mention of purple is not accepted. Students will also be penalized if they wrote that CO_2 gas was produced as it would not be the case with **A**.

A few students wrote "heat under reflux". Students are to take note that any distinguishing test carried out is usually a **simple** chemical test that is conducted using test-tubes. The reflux condenser should not be used as, firstly, one cannot easily connect the test-tube to a reflux condenser. Secondly, the reaction set-up would be too elaborate and not meet the requirements of "a simple chemical test".

- 2 (b) (i) **(I)** is more stable due to the delocalization of π electrons around the ring. [1]

The understanding of resonance effect in a benzene ring is required for this question. Students have to demonstrate the understanding of having π electrons delocalised in the π orbitals. Merely stating that actual benzene has resonance effect or the p orbitals are aligned is insufficient.



energy level diagram with correct labels [1]
correct value for enthalpy change of hydrogenation [1]

Students are to piece together the equations and information given to them in order to construct the energy level diagram. Since the actual structure of benzene is 149 kJ mol^{-1} more stable than the hypothetical structure of benzene, benzene should be lower in energy than the hypothetical structure. Enthalpy change of hydrogenation of one $\text{C}=\text{C}$ bond in cyclohexene is given to be -118 kJ mol^{-1} . Thus the enthalpy change of hydrogenation of three $\text{C}=\text{C}$ in the hypothetical benzene structure would be three times of (-118) . Combining this two information together, the energy level diagram can be constructed and the missing enthalpy change of hydrogenation of benzene can be calculated.

ECF is awarded for correct calculations made with a complete but incorrect energy level diagram. Students are to take note that the direction of the arrows

in the energy level diagrams and the sign of the enthalpy change has to correspond to each other. i.e. an arrow pointing upwards should be a positive enthalpy change, vice versa.

Mrs Lim XT: S60 – S6H; S70 – S75; S7F – S7J

Ms Fan YT: S6J – 6Q; S76 – S7E

3 (a) (i) $K_p = \frac{(P_{\text{C}_7\text{H}_8})(P_{\text{H}_2})^4}{P_{\text{C}_7\text{H}_{16}}}$ [1]

Students should recognise that the equilibrium constant K_p is to be written in terms of partial pressure and not in terms of concentration. No mark is awarded if any square bracket [] is used in the expression, as this is referring to the concentration of a substance.

3 (a) (ii)

	$\text{C}_7\text{H}_{16}(\text{g})$	$=$	$\text{C}_7\text{H}_8(\text{g})$	$+$	$4\text{H}_2(\text{g})$	
Initial / mol	10		0		0	
Change / mol	-x		x		4x	
Equilibrium / mol	10-x		x		8	
	= 8		= 2			x = 2 [1]
						total n = 18
Partial pressure / atm	(8/18)(3)		(2/18)(3)		(8/18)(3)	
	= 1.333		= 0.3333		= 1.333	[1]

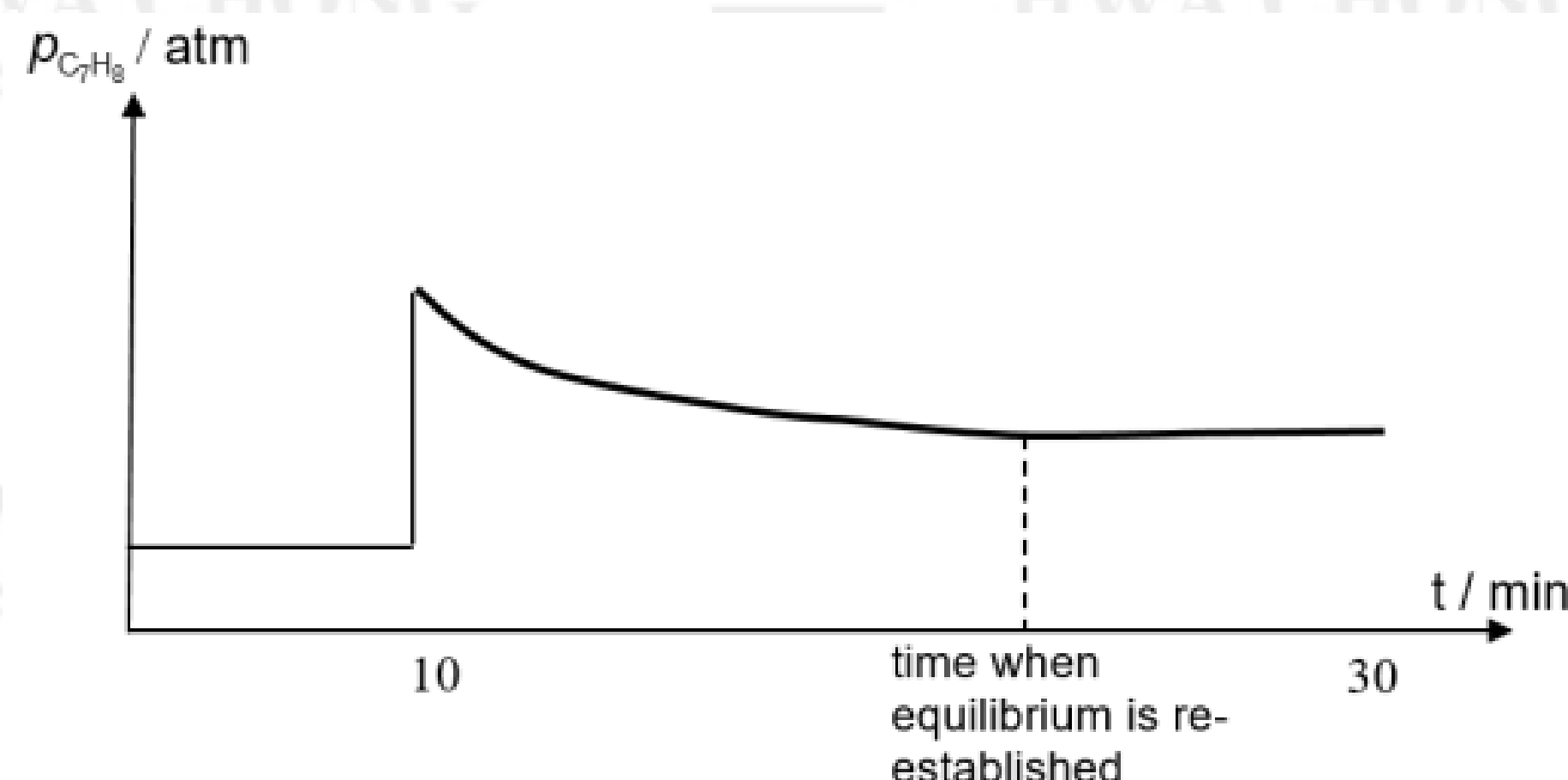
$$K_p = (0.3333)(1.333)^4 / 1.333 = 0.790 \text{ atm}^4 \quad [1]$$

This chemical equilibrium calculation is generally well done. The question provided us with the initial amount of C_7H_{16} . An ICE table should then be used to determine the equilibrium amount of the reactant and products. In order to obtain K_p , students are required to substitute the equilibrium partial pressures into the K_p expression, and they need to be aware that partial pressure = mole fraction $\times$ total pressure. The last mark is not awarded if candidates miss out / gave a wrong unit.

- 3 (b) (i) When a smaller vessel is used, the total pressure increases. By Le Chatelier's principle, position of equilibrium shifts to the left so as to decrease the pressure by favouring the production of fewer number of moles of gases, hence the yield of methylbenzene decreases. [1]

Students are required to provide all the underlined points to gain full credit for this question. $\frac{1}{2}$ mark is deducted for any missing point. Students need to mention that the position of equilibrium is shifted to the left due to the "fewer moles of gas" on the left side of the equation.

3 (b) (ii)



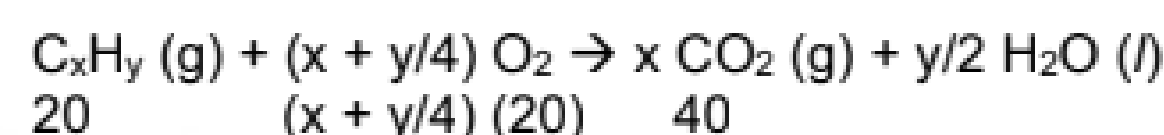
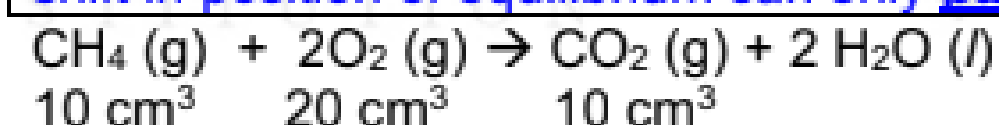
Marking points: p decreases from 10 min until when eqm re-established, following which the line becomes horizontal [½]; new eqm p higher than old eqm p [½]

When the partial pressure of C_7H_8 is suddenly increased at $t = 10$ min, the position of equilibrium would shift to the left to partially offset this increase. The partial pressure of C_7H_8 would thus decrease until the point when equilibrium is re-established.

When equilibrium is re-established, the rate of the forward reaction is equal to the rate of the backward reaction. There is thus no further change in the partial pressure of C_7H_8 . This is thus indicated by a horizontal line on the graph.

Students need to also recognise that the partial pressure of the C_7H_8 at the new equilibrium position is higher than that at the old equilibrium position, as the shift in position of equilibrium can only partially offset the disturbance.

3 (c)



Volume of CO_2 reacted with C_xH_y : $70 - 20 = 50\text{ cm}^3$

$$\frac{n_{\text{CO}_2}}{n_{\text{C}_x\text{H}_y}} = \frac{40}{20} = \frac{x}{1}$$

$x = 2$ [1]

Volume of O_2 left: $100 - 20 - (20)(2 + y/4) = 40 - 5y = 20\text{ cm}^3$
 $y = 4$ [1]

This question proves to be challenging for quite a number of students. Students would not be given full credit if their working is unclear. No credit is given if a student happens to get $x = 2$ or $y = 4$, but via a wrong method.

In this question, both CH_4 and C_xH_y undergoes complete combustion to form CO_2 and H_2O . CO_2 is an acidic gas which reacts with the $\text{Ca}(\text{OH})_2$ provided, resulting in a decrease of 50 cm^3 . Based on Avogadro's Law, the combustion of 10 cm^3 of CH_4 would produce 10 cm^3 of CO_2 . This implies that 40 cm^3 of CO_2 is produced from the combustion of C_xH_y . Since the combustion of 20 cm^3 of C_xH_y produces 40 cm^3 of CO_2 , $\frac{n_{\text{CO}_2}}{n_{\text{C}_x\text{H}_y}} = \frac{40}{20} = \frac{x}{1}$ and thus $x = 2$.

The question mentions that "After cooling to room temperature, 70 cm^3 of the gas mixture remained". This 70 cm^3 of gas is thus made up of excess O_2 and CO_2 . There is no $\text{H}_2\text{O}(\text{g})$ in this 70 cm^3 as it would have condensed to form liquid water after cooling to room temperature. After the 50 cm^3 of CO_2 has been removed by $\text{Ca}(\text{OH})_2$, the 20 cm^3 that remains is actually excess oxygen, and not H_2O . This implies that 80 cm^3 of O_2 has been used for the combustion of CH_4 and C_xH_y . Based on Avogadro's law, 20 cm^3 of O_2 is needed to react with 10 cm^3 of CH_4 and $(x + y/4) \times (20)\text{ cm}^3$ of O_2 is needed to react with 20 cm^3 of C_xH_y . This allows us to solve for y .

Mrs Phee PL: S60 – S6Q; S70; S7J

Mr Low KS: S71 – S79; S7A – S7G

4 (a) To keep the total volume constant so that the concentration of propene or bromine used is directly proportional to the volume used. [1]

Conc. of reactant in reaction mixture = (Original conc. of reactant $\times$ volume of reactant used) $\div$ Total volume of mixture

By keeping the total volume constant at 40 cm^3 for all the experiments, we can take [reactant] $\propto$ volume of reactant used, which makes comparison of data easier. This part should be included in the answer but was often excluded.

4 (b) Rate of reaction is inversely proportional to the time taken for the colour of bromine to disappear. [1]

In general, when the time for the reaction to be completed is shorter, the rate of reaction is faster. Hence, rate is inversely proportional to time.

4 (c) Comparing experiments 1 and 2,

when volume (and hence concentration) of propene solution used is decreased 0.6 times, the rate decreases 0.6 times as well.

Since [propene] is directly proportional to rate, it is first order with respect to propene.

1st order wrt propene [1] + explanation [1]

Experiments 1 and 2 are used for comparison since the [bromine] is the same and thus any change in rate in these experiments must be solely due to the change in [propene].

4 (d) Comparing experiments 1 and 3,

rate for Expt 1 $\propto$ 20/18

rate for Expt 3 $\propto$ 10/18

Let order for Br_2 be y .

$$(20/10)^y = (20/18) / (10/18)$$

$$y = 1$$

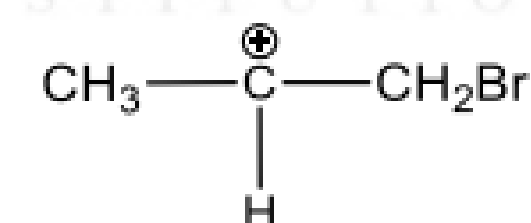
explanation [1]

When volume and hence $[\text{Br}_2]$ is halved, the number of moles of bromine to be completely reacted to cause the colour to disappear is also halved. Thus the time needed should be halved as well if both reactions occur at the same rate.

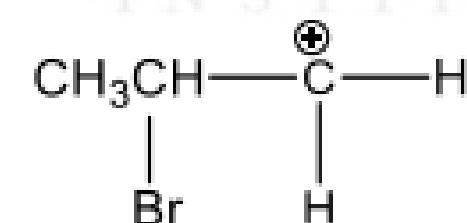
However, it is noted that the time taken for both experiments 1 and 3 is the same, implying that the rate of reaction for Experiment 3 must be slower (halved).

Hence, when $[\text{Br}_2]$ is halved, rate of reaction is halved. $\text{Rate} \propto [\text{Br}_2]$. Thus it must be first order with respect to Br_2 .

4 (e)



intermediate leading to X



intermediate leading to Y

There are more electron donating alkyl groups attached to the positively charged carbon in the first carbocation which is therefore more stable. The first carbocation then gives **X** as the main organic product.

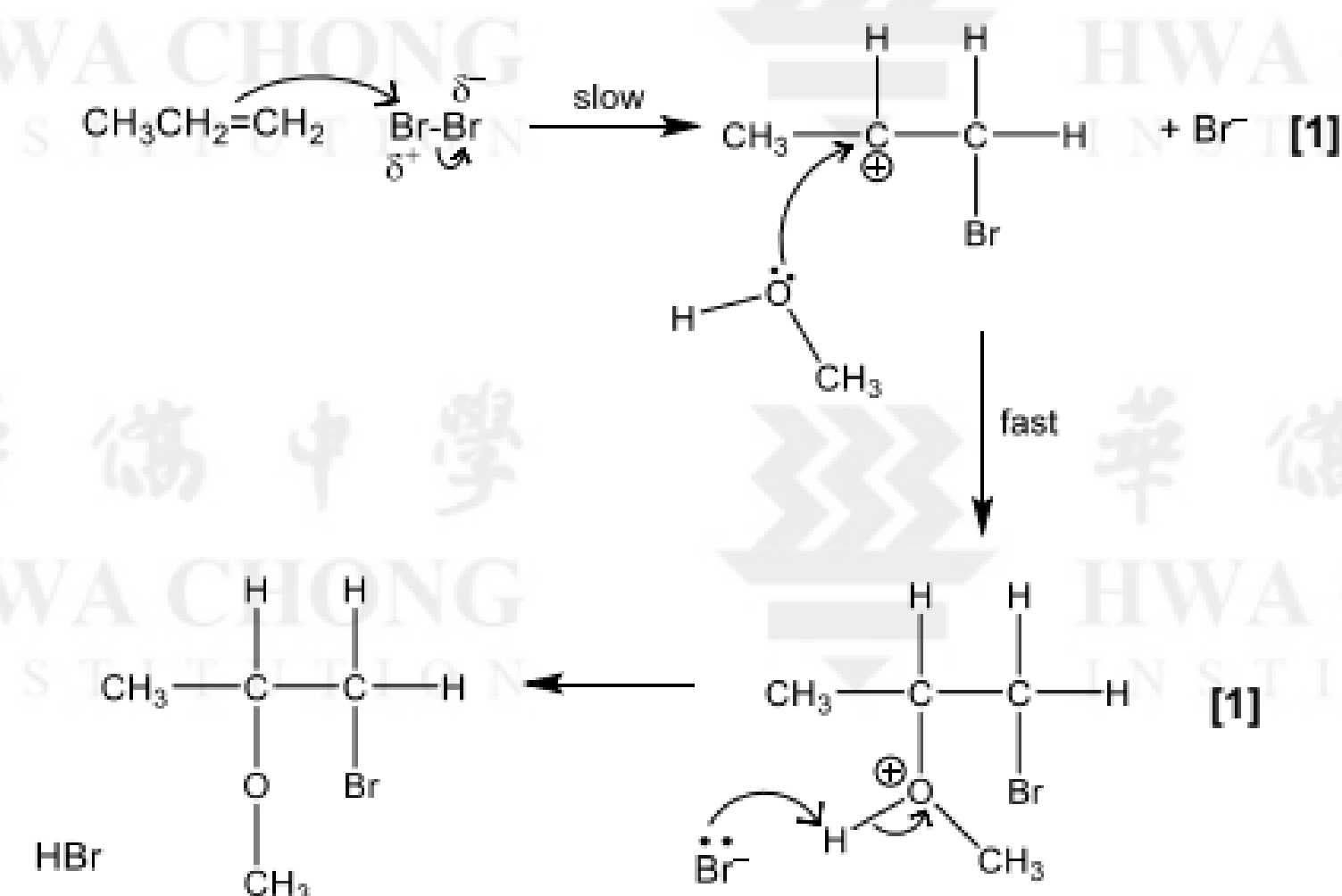
[1] draw structures of two possible carbocations (no $\frac{1}{2}$ mark awarded)

[1] explanation

In the first step of the electrophilic addition with Br_2 , one of the bromine atoms of Br_2 will be attached to either of the carbon atoms of the $\text{C}=\text{C}$ of propene, leading to the two carbocations as shown above.

Alkyl groups are electron donating groups which help to disperse the positive charge on the positively charged carbon in the carbocation. Recall from your lecture notes that in general, organic species with a charge on it are less stable. Therefore, the carbocation leading to **X** has more electron donating groups which means that carbocation is more stable.

4 (f) Electrophilic addition [1]



[-1/2 for any mistake]

The first step involves the formation of the carbocation. After that, methanol which has available lone pairs will be able to attack the carbocation to form an intermediate. A further deprotonation to generate the final product will then occur.

Do be careful when drawing the curly arrows, e.g.: the arrow should start from the $\text{C}=\text{C}$ bond and point towards the δ^+ Br; the arrow should start from the lone pair of electrons on the O of methanol and point towards the C carrying the positive charge (**not** point at the positive charge).

Paper 3

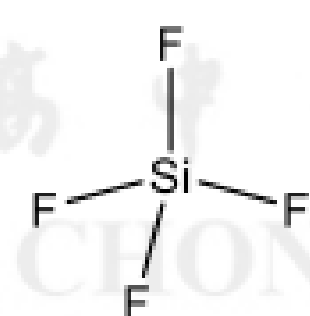
Ms Wong HM: S60; S6N – S6Q; S70 – S76

Mrs Lee CK: S61 – S69

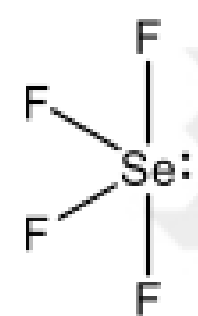
Mrs Lim SF: S6C – S6K; S7J

Mr Loy HC: S77 – S7G

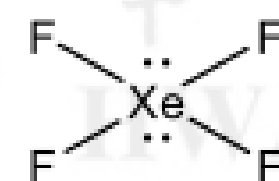
1 (a)



tetrahedral



see-saw



square planar

For each molecule,

[½] name of the shape

[½] diagram to illustrate the shape

Si has 4 valence electrons, F forms single bonds. So SiF₄ has 4 bond pairs and 0 lone pair around the central atom, ∴ SiF₄ is tetrahedral.

SeF₄ and XeF₄ also have 4 bond pairs each (4 Se–F bonds / 4 Xe–F bonds respectively) but Se has 6 valence electrons, Xe has 8.

∴ SeF₄: 4 bond pairs and 1 lone pair, ∴ see-saw shape.

XeF₄: 4 bond pairs and 2 lone pairs, ∴ square planar.

Some students lost marks ∴ their diagrams do not look like the correct shape (e.g. a diagram that does not look tetrahedral)

Dot-and-cross diagram is **not** used to **illustrate** shapes of molecules.

1 (b) (i) [½] increase in pressure pushes gas molecules closer together or more gas molecules per unit volume (must mention 'molecules')

[½] increase in frequency of collision between reactant molecules

It is the same as asking you to explain why an increase in concentration would increase the rate of a reaction.

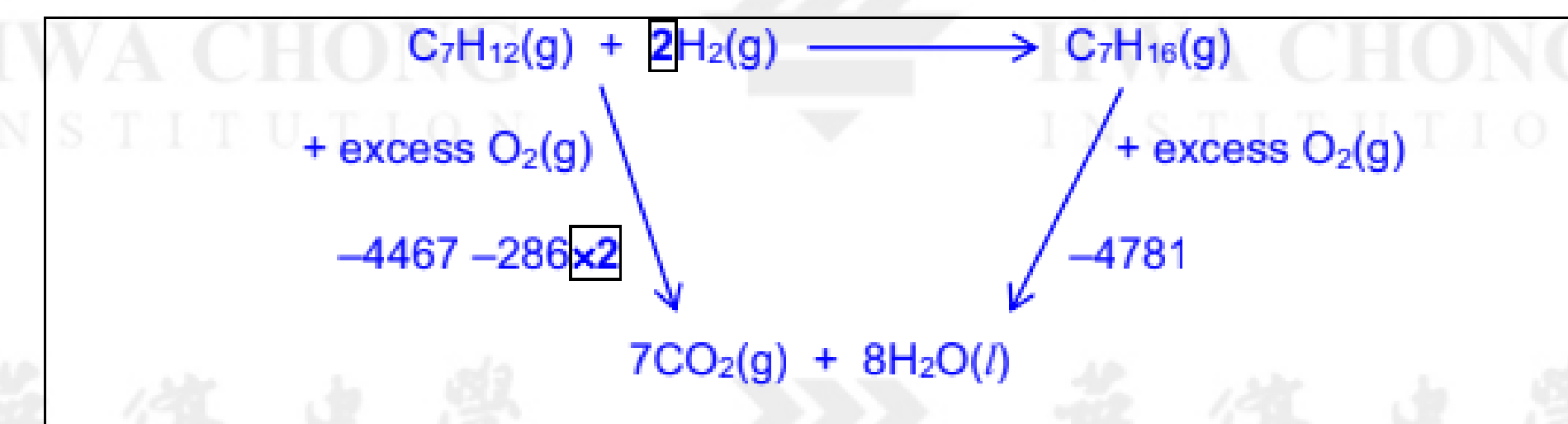
It is **incorrect** to discuss 'position of equilibrium' ∴ that concerns the *yield* of the product at equilibrium; anyway there is no equilibrium here.

It is **incorrect** to mention 'increase in kinetic energy of the molecules' ∴ the temperature of the system did not change.

1 (b) (ii) [2] $\Delta H = -4467 - 286 \times 2 - (-4781) = -258 \text{ kJ mol}^{-1}$

This question concerns ΔH calculations involving Hess' law and ΔH_c . If you forget whether it is 'reactants – products' or 'products – reactants', you may draw a simplified energy cycle to decide how the three ΔH_c values can be combined to determine ΔH .

$\Delta H = ?$



1 (b) (iii) [1] Hydrogenation of alkenes involve high activation energy / catalyst was added to lower the activation energy of the reaction.

The question says the reaction is 'thermodynamically favourable' i.e. thermodynamically spontaneous i.e. ΔG is negative, but 'will not proceed without catalyst'. So this must be the problem of E_a being too high.

Some students attempted to explain why E_a was high in terms of a "weak hydrogen *electrophile*". This is **incorrect** as the hydrogenation of alkenes does **not** involve formation of electrophiles. Instead, the mechanism occurs via adsorption/desorption of reactants/products on the catalyst's surface.

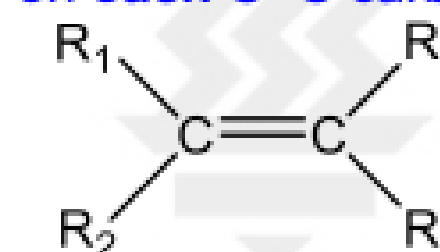
Some students wrote answers like 'to overcome the activation energy' – this was **not** accepted ∴ this phrase does **not** suggest that the 'original' E_a was high. The phrase 'to overcome the activation energy' is also **not** appropriate here ∴ when the catalyst is added, you would have a 'new' E_a .

1 (c) (i) [1] there is restricted rotation about the C=C double bond or restricted rotation due to the π bond of the C=C double bond

[1] there must be two different groups attached to each carbon of the C=C bond

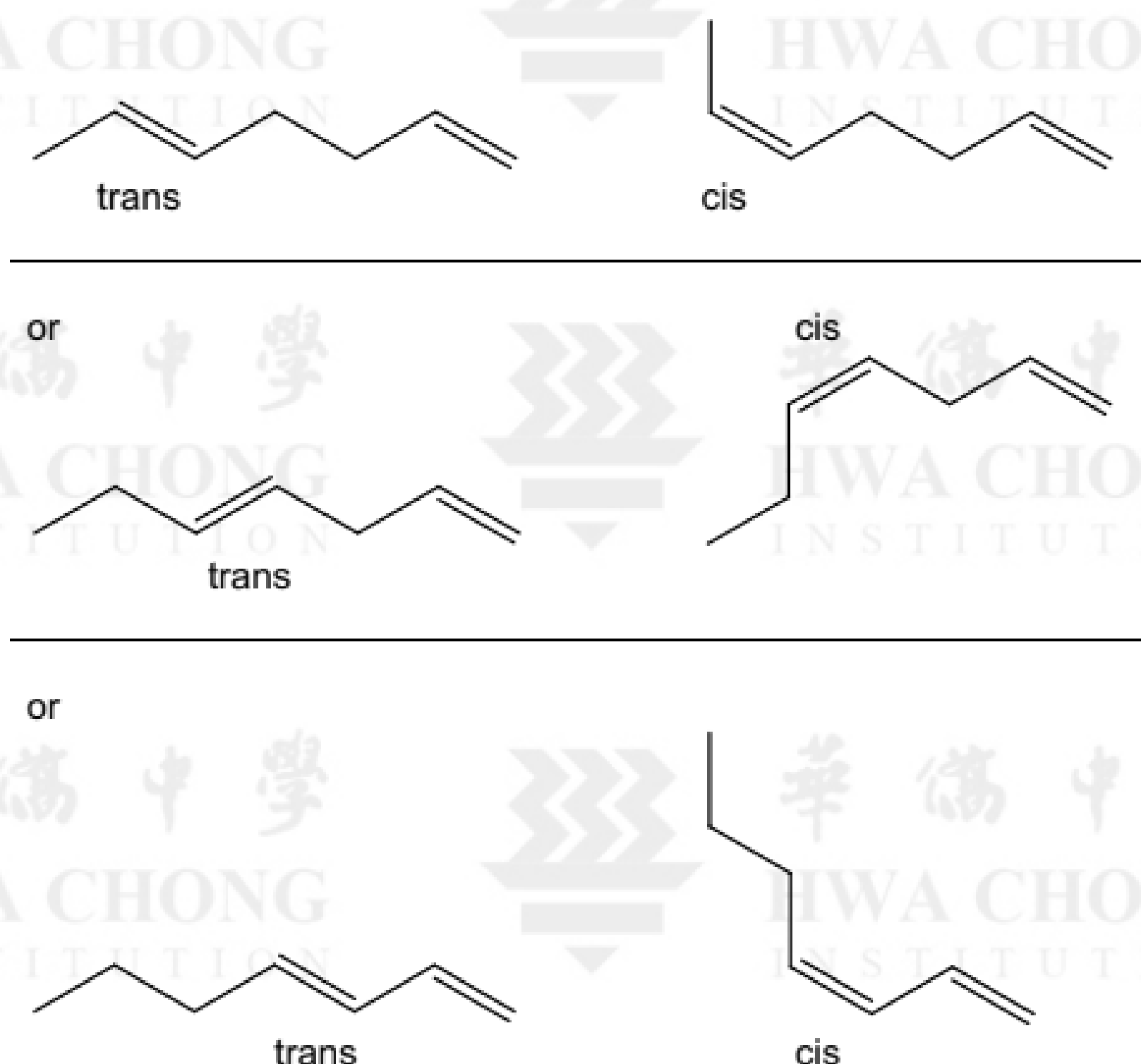
Some students used terms like 'two atoms' / 'two molecules' / 'two compounds' / 'two constituents' – these are **incorrect**.

Some students used terms like 'same side of the C=C' / 'different side' / 'each side of the C=C' – these were **not** accepted ∴ are you referring to R₁ vs. R₃ and R₂ vs. R₄ or R₁ vs. R₂ and R₃ vs. R₄? Cis-trans isomerism should arise due to different groups on **each** C=C carbon, i.e. R₁ vs. R₂ and R₃ vs. R₄.

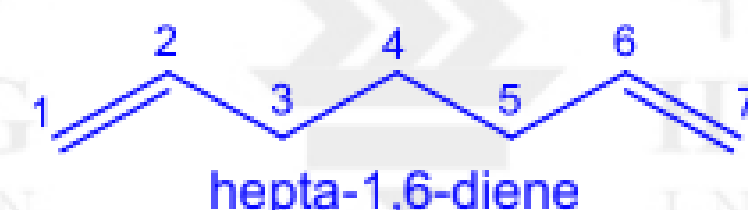


Some students used statements that sound that R₁ R₂ R₃ R₄ must all be different – this is **incorrect**.

1 (c) (ii) [1] + [1] draw and label the 'cis' and 'trans' isomers for compound P



The question gave hepta-1,6-diene. Based on understanding of the answers in (c)(i), hepta-1,6-diene does not have cis-trans isomers.



The question wants an **unbranched non-cyclic** constitutional isomer that has only two cis-trans isomers $\Rightarrow$ a simple answer is to change the position of one of the alkene groups in hepta-1,6-diene from $C^1=C^2$ to $C^2=C^3$. It is alright to (follow the question and) use skeletal formula to draw these cis-trans isomers. In fact if you use skeletal formula, you have already drawn out the trans isomer.

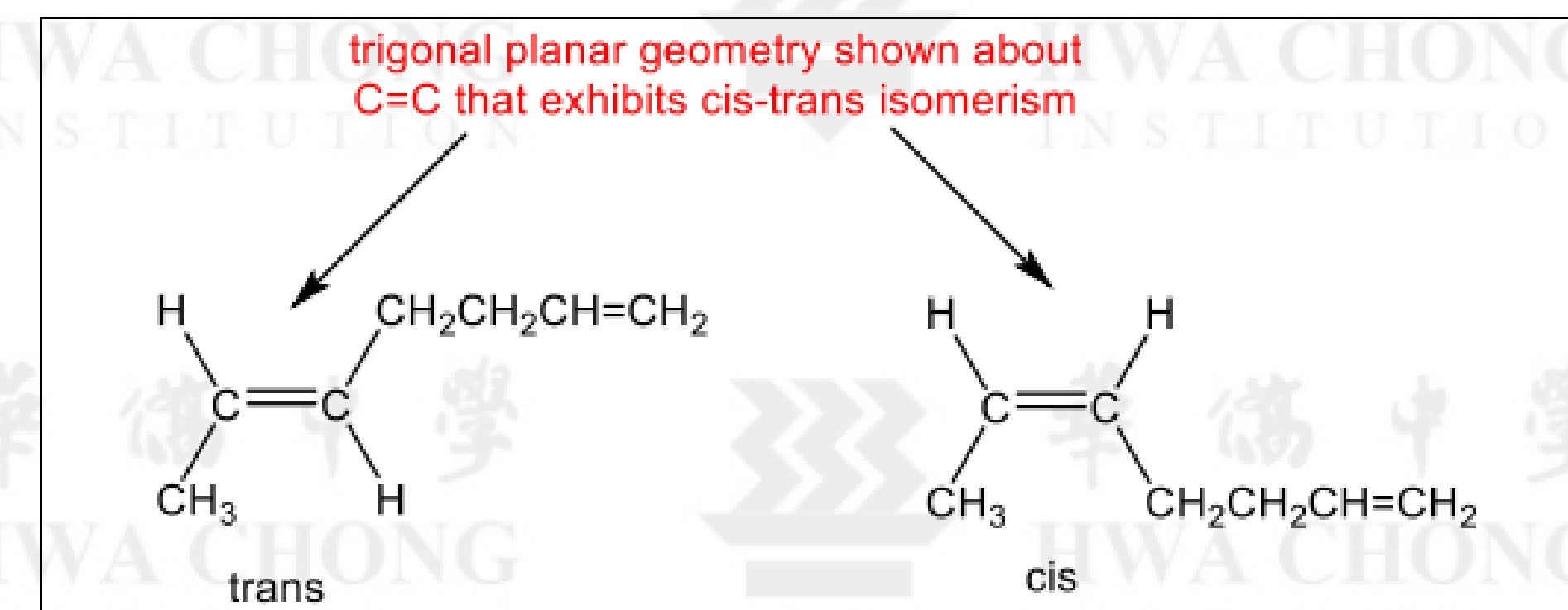
Some students drew structures that would have more than two cis-trans isomers $\Rightarrow$ 0 mark

Some students used 'molecular formula' like ' C_4H_7 ' which is ambiguous, the $C=C$ could be at a position such that the diene has more than two cis-trans isomers $\Rightarrow$ 0 mark

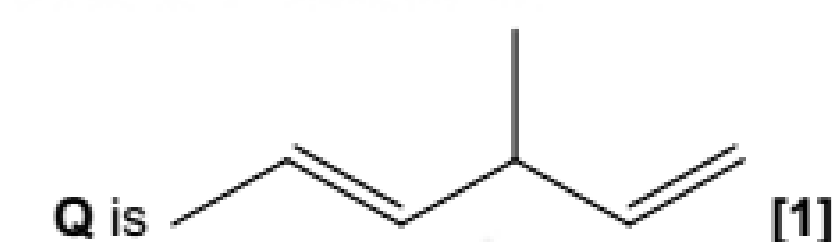
Some students drew structures with one missing $C=C \Rightarrow$ 0 mark

Some students drew structures with extra or missing carbon $\Rightarrow$ penalised

Note that if you chose to represent cis-trans isomers using displayed formula or condensed structural formula, trigonal planar geometry must be shown for the $C=C$ which exhibits cis-trans isomerism. An example is given below.



1 (c) (iii)



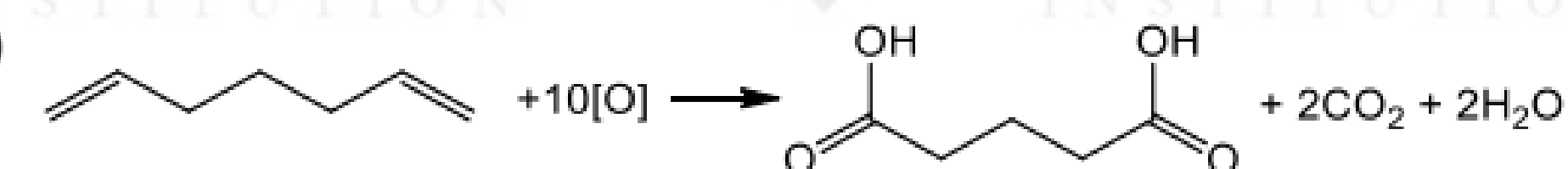
The question gave hepta-1,6-diene which does not have cis-trans isomers nor enantiomers.



We want a non-cyclic constitutional isomer **Q** with cis-trans isomers and enantiomers. So at least one of the alkene groups cannot be a 'terminal' $=CH_2$ and you need to rearrange the 'carbon framework' to make a chiral carbon, which means **Q** must have a branched structure.

Some students used 'molecular formula' like ' C_3H_5 ' which is ambiguous, the $C=C$ could be at a position such that there is no cis-trans $\Rightarrow$ 0 mark

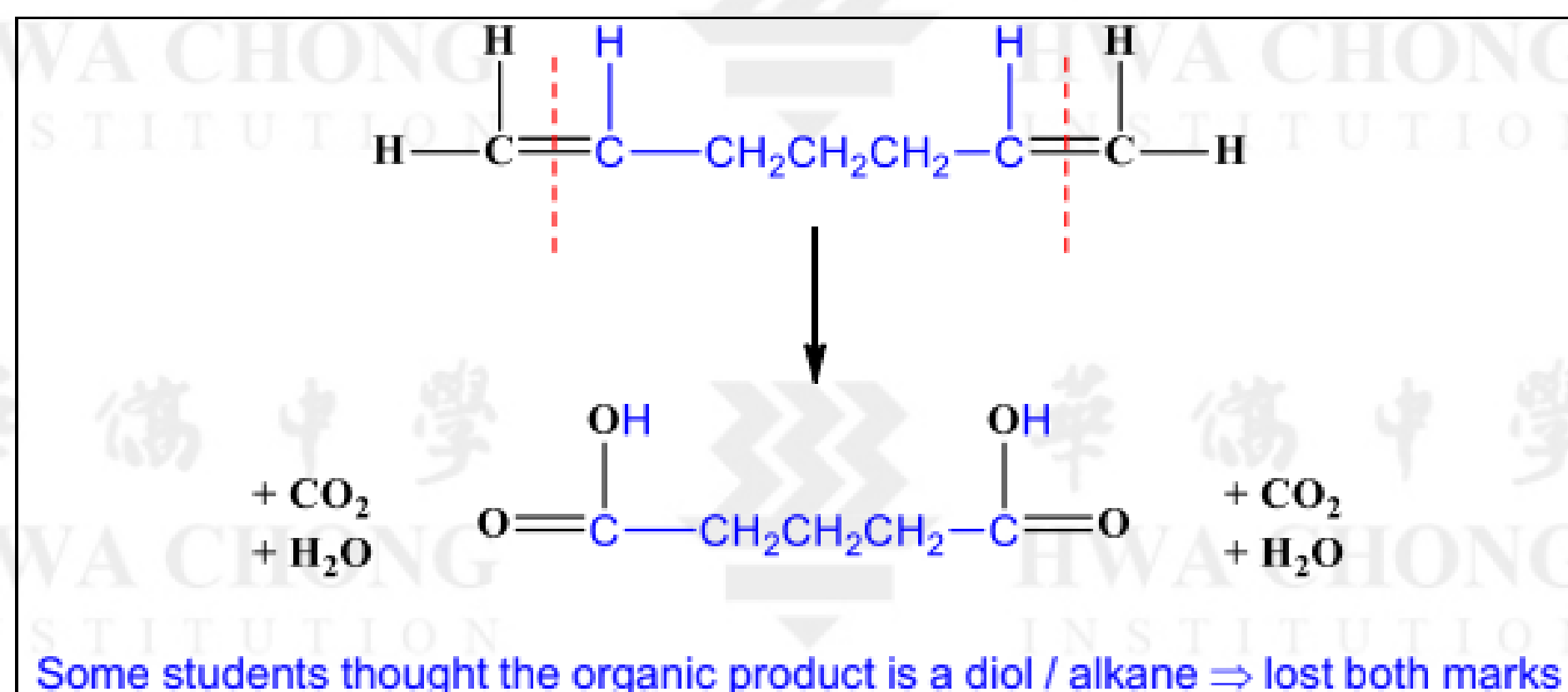
1 (c) (iv)



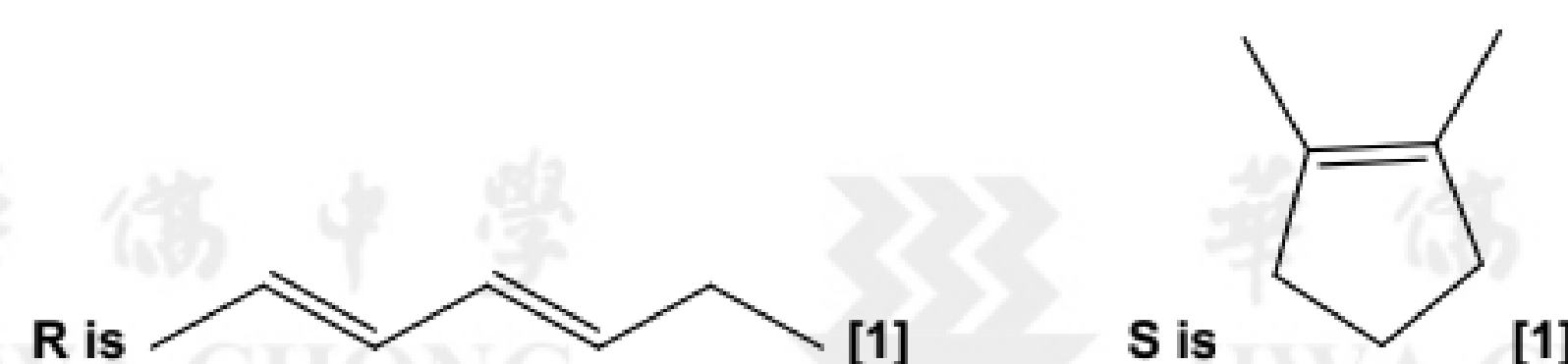
[1] structure of organic product

[1] equation balanced with CO_2 H_2O [O]

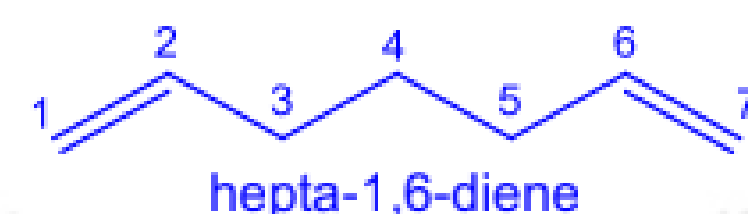
Most students knew oxidative cleavage would occur and gave the correct carbon-containing products (dicarboxylic acid and $2CO_2$). Most students then correctly filled in $2H_2O$ to balance the H, they then correctly filled in $10[O]$ to balance the O.



1 (c) (v)



The question gave hepta-1,6-diene C_7H_{12}



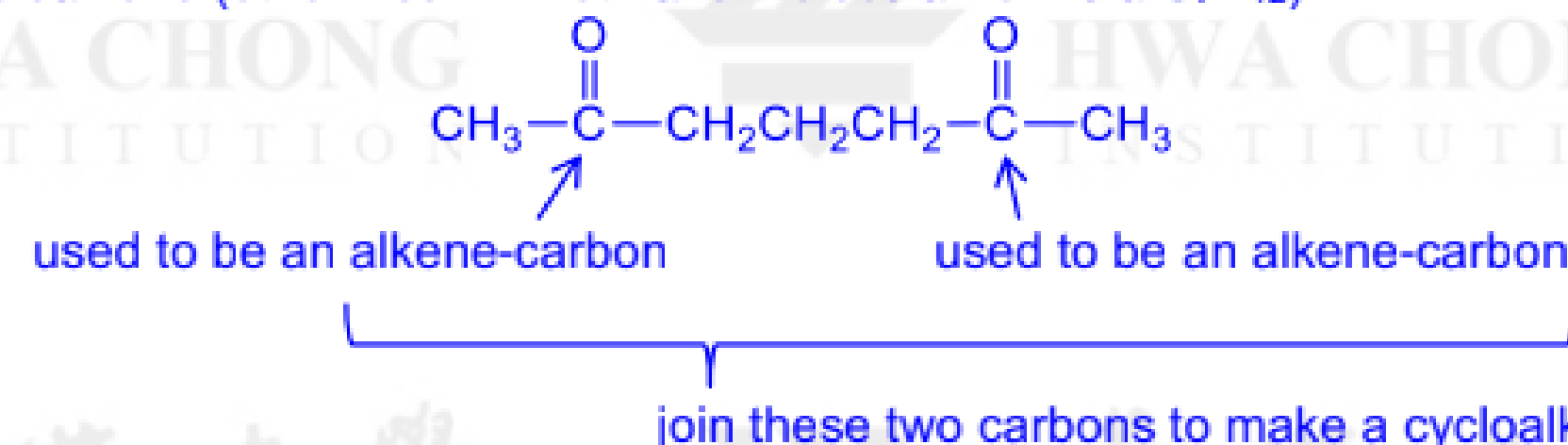
R is unbranched, non-cyclic.

R has four cis-trans isomers, $\therefore$ both alkene groups are not 'terminal' $=CH_2$. (If **R** has two 'terminal' $=CH_2$, it would be the given hepta-1,6-diene!)

1 mol **R** gives 2 mol CO_2 after oxidative cleavage, but both alkene groups are not 'terminal' $=CH_2$, $\therefore$ the two alkene groups must be of the form $C=CHCH=C$.

Hence draw **R** (just change the positions of the $C=C$ from the given diene).

The oxidative cleavage product from **S** still has 7 carbons, $\therefore$ **S** must be a cycloalkene (otherwise will not have molecular formula C_7H_{12}).



Double-check whether your proposed **R** and **S** are correct – let them undergo oxidative cleavage and see that they do give the specified products.

1 (d)

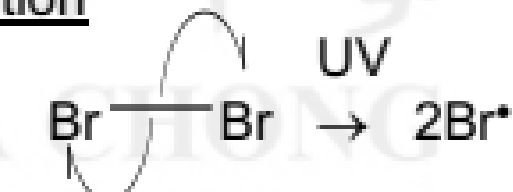
[1] free radical substitution

[1] initiation, equation, fish-hook arrows
(three marking points, $-\frac{1}{2}$ for each missing item / error)

[1] propagation, two equations
(three marking points, $-\frac{1}{2}$ for each missing item / error)

[1] termination, at least two equations
(three marking points, $-\frac{1}{2}$ for each missing item / error)

Initiation



Propagation



Termination



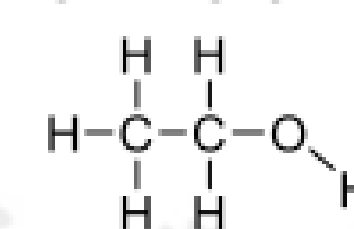
This question is book-work. The question advised you to represent heptane as RCH_3 and 1-bromoheptane as RCH_2Br .
Some students were penalised for labelling the $\cdot$ on the R.

Mr Lee GH: S60 – S69; S70

Mr C Loy: S6C – S6Q; S71

Dr B Chan: S72 – S7J

2 (a) (i)



Correct displayed formula [1]

A 'displayed formula' requires all covalent bonds to be shown, including the $-O-H$ bond.

2 (a) (ii) Angle of $H-C-H$: 109.5° [$\frac{1}{2}$]

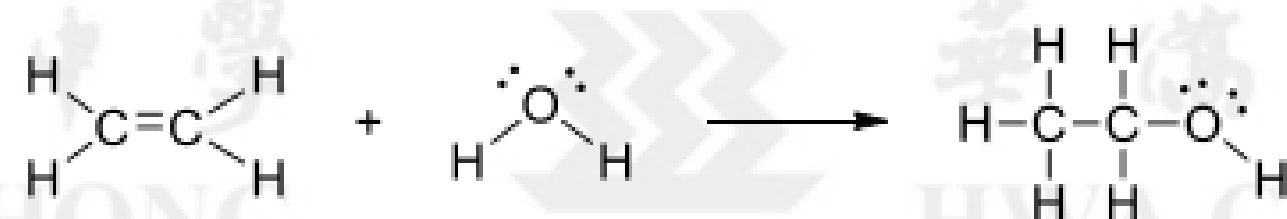
Angle of $C-O-H$: 104.5° [$\frac{1}{2}$]

Since there are no lone pairs around carbon/four bond pairs around carbon but there are two lone pairs around oxygen [1] and since lone pair-lone pair repulsion > lone pair-bond-pair repulsion > bond pair-bond pair repulsion, the bond angle of $C-O-H$ in ethanol is smaller. [1]

Note the correct way to present your answers:

- Choice of words: 'the number of lone pairs on the O atom' (**not** 'the number of lone pairs around the C–O–H group')
- Use of abbreviations like bp, lp, etc: must write in full at least once, e.g. write in full when mentioning the first time in your answer, and after that you may use the abbreviations.

2 (a) (iii)



$$\begin{aligned}\Delta H &= \sum \text{B.E. (bonds broken)} - \sum \text{B.E. (bonds formed)} \\ &= [610 + 4(410) + 2(460)] - [350 + 5(410) + 360 + 460] \\ &= \underline{-50 \text{ kJ mol}^{-1}}\end{aligned}$$

[1] all bond energies correct and multiplied by correct coefficients

[1] final answer (no ecf)

Always consider the structure of all the reactants and products so that you don't miss out any bonds. Note that bonds broken always consume energy (so the bond energy value has positive sign), and bonds formed always release energy (so a negative sign is placed in front of the bond energy value).

2 (a) (iv) The value of equilibrium constant will increase since a lower temperature will favour the forward exothermic reaction which will generate additional heat. Hence, the position of equilibrium will shift to the right, favouring the formation of products, hence increasing the ratio of concentration of products to that of reactants.

[1] equilibrium constant increase (no mark for guessing)

[1] correct reasoning: exothermic reaction/generate heat **AND** favour forward reaction/POE shifts right

The effect of a decrease in temperature (from 300 °C to 150 °C) is to cause the equilibrium to shift in the direction that opposes the change, i.e. in the direction of the forward exothermic reaction. This results in an increase in the proportion of products to reactants (you can call this 'an increase in the extent of reaction'), and hence the equilibrium constant increases.

Your answer must clearly state how K_c changes, followed by an explanation along the lines of 'the position of equilibrium shifts (to which side) to (achieve what result)'.

For a reversible reaction, you must be careful that a change in temperature will affect the rates of both the forward and reverse reactions. In this case, since the temperature is decreased, you cannot say 'there is an increase in the rate of the forward reaction'. In fact, when temperature is decreased, both forward and reverse rates decrease (to different extents), such that the position of equilibrium shifts to the right.

2 (a) (v) 1. concentrated H_2SO_4 , cold
2. water [1]

You have given us many incorrect versions of this answer.

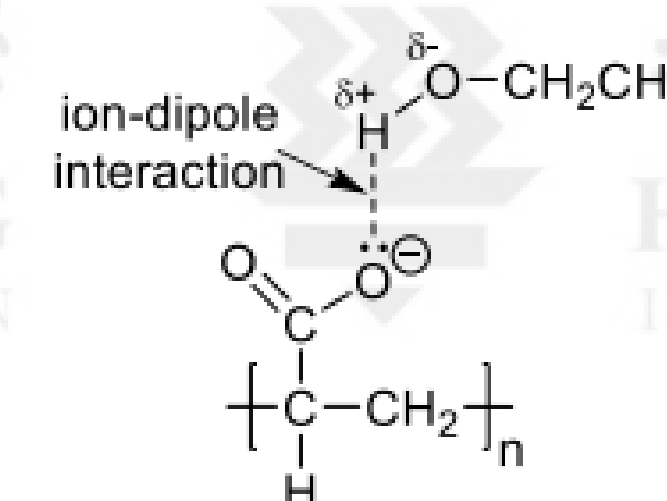
For this laboratory synthesis, the reagents and conditions should be given in a sequence of two steps:

Step 1: must be concentrated and cold sulfuric acid (**not** dilute or aqueous acid, and **not** any other acids)

Step 2: add water, or followed by (addition of) water (no need to mention 'room temperature' or 'cold', and definitely **not** 'heat / warm / steam / $\text{H}_2\text{O(g)}$ ').

Students need to be aware that conc H_2SO_4 consists largely of H_2SO_4 molecules which are un-ionised because there are not enough water molecules to bring about the formation of aqueous ions. Hence, it is written as conc $\text{H}_2\text{SO}_4(l)$. In contrast, dilute H_2SO_4 is completely ionised, and is written as $\text{H}_2\text{SO}_4(aq)$.

2 (b)



[1] label ion-dipole interaction / hydrogen bonding

[1] Diagram. Deduct ½ mark for every mistake (wrong structure of carboxylate ion/ethanol; missing charge on carboxylate ion; missing δ^+ and δ^- , missing lone pair (if the answer is hydrogen bond)).

Since the polymer has an anion group, the better answer here is 'ion-dipole interaction'. In this case, your diagram should show the dipole on the O-H bond in $\text{C}_2\text{H}_5\text{OH}$ interacting with the O^- in the -CO_2^- group (as drawn above). No need to show any lone pair on the O^- ion, or dipole on any part of the -CO_2^- group.

If your answer is 'hydrogen bond', we expect to see O-H bond in $\text{C}_2\text{H}_5\text{OH}$ interacting with either the lone pair on the O^- ion or on the doubly bonded O atom in the -CO_2^- group. No need to show any other lone pair or dipole anywhere else.

2 (c) (i) **Ethanol and water molecules form stronger hydrogen bonding** with each other rather than that between ethanol molecules and that between water molecules. [1]

Your explanation for 'shrink in volume' must include a comparison of the intermolecular forces in the ethanol-water mixture (hydrogen bonds) with those in the ethanol and water (also hydrogen bonds).

- 2 (c) (ii) Mass of ethanol in 200 cm³ of ethanol-water mixture = 140 × 0.789 = 110.46 g
 Mass of water in 200 cm³ of ethanol-water mixture = 60 × 0.998 = 59.88 g
 Both masses correct [1]

$$n(\text{ethanol}) \text{ in } 200 \text{ cm}^3 \text{ of ethanol-water mixture} = \frac{110.46}{2(12.0) + 6(1.0) + 1(16.0)} = 2.40 \text{ mol}$$

$$n(\text{water}) \text{ in } 200 \text{ cm}^3 \text{ of ethanol-water mixture} = \frac{59.88}{2(1.0) + 1(16.0)} = 3.33 \text{ mol}$$

$$\text{mole fraction of ethanol } (x_1) = \frac{2.40}{2.40 + 3.33} = 0.419 \text{ [1] ecf based on wrong masses}$$

The units of 'mol' is always required for amounts of substances; the correct format is: $n_x = \text{___ mol}$, or $n(X) = \text{___ mol}$.
 Intermediate answers can be reported to 4 or 3 s.f., while final answers must be 3 s.f. (unless otherwise stated in the question).

- 2 (c) (iii) From the graph, excess molar volume is $-1.08 \text{ cm}^3 \text{ mol}^{-1}$ when $x_1 = 0.420$
 $\Rightarrow$ volume contraction of 200 cm³ ethanol-water mixture = $1.08 \times (2.40 + 3.33)$
 $= 6.19 \text{ cm}^3$ [1] ecf

The graph only gives 'excess molar volume', so the shrink in total volume requires one more step: excess molar volume multiplied by total amount in mol.

Mrs J Lim: All classes

- 3 (a) (i) Since volume of $\text{S}_2\text{O}_3^{2-}$ is directly proportional to $[\text{I}_2]$, when volume of $\text{S}_2\text{O}_3^{2-}$ and hence $[\text{I}_2]$ decreases, gradient which is the rate of the reaction is constant. Rate is independent of $[\text{I}_2]$, thus order of reaction with respect to I_2 is zero. [1]

When [propanone] increases 2 times, gradient of expt 2/ expt 1 = 2 (show calculation) [1], rate also increases by 2 times, rate is directly proportional to [propanone], thus order of reaction with respect to propanone is one. [1]

I_2 (in the withdrawn sample) was titrated with $\text{S}_2\text{O}_3^{2-}$. Therefore, $[\text{I}_2]$ remaining in the reaction mixture is directly proportional to **Volume $\text{S}_2\text{O}_3^{2-}$** . Therefore, the given graph is equivalent to a graph of $[\text{I}_2]$ vs. time and the gradient of this graph represents the rate of reaction. We have a straight line (for either graph) with constant gradient, so rate stays constant even though $[\text{I}_2]$ is changing. Hence the reaction is independent of $[\text{I}_2]$ i.e. zero order for I_2 .

To determine the order with respect to propanone, we have to compare the two graphs i.e. compare the gradient for the two lines – the only difference between experiment 1 and experiment 2 is that [propanone] was varied.

- 3 (a) (ii) Rate = $k[\text{H}^+][\text{propanone}]$ [1] ecf from orders found

Rate equation is expressed as: **Rate = $k [\text{A}]^m [\text{B}]^n$** ..., where m and n represent the order of reaction with respect to reactants A and B respectively. Since question mentioned that the order of reaction with respect to H^+ is 1,

students are expected to combine this information together with the answer in (a)(i) to come out with the overall rate equation.

- 3 (a) (iii) Gradient for experiment 1 = $(16.4 - 10.4) / (0 - 20) = -0.300 \text{ cm}^3 \text{ min}^{-1}$ [½]

$$\text{Change in no. of moles of } \text{I}_2 / \text{min} = \frac{1}{2} (0.0500)(0.300 \times 10^{-3})$$

$$= 7.50 \times 10^{-6} \text{ mol min}^{-1} \text{ [½] ecf on gradient}$$

Since number of moles of I_2 can be obtained from number of moles of $\text{S}_2\text{O}_3^{2-}$ from the second equations given: $2\text{S}_2\text{O}_3^{2-} + \text{I}_2 \rightarrow \text{S}_4\text{O}_6^{2-} + 2\text{I}^-$, change in number of moles of I_2 per minute can be found from change in number of moles of $\text{S}_2\text{O}_3^{2-}$ per minute. To find the change in number of moles of $\text{S}_2\text{O}_3^{2-}$ per minute, we can get it from the gradient of the graph, which represents the change in volume of $\text{S}_2\text{O}_3^{2-}$ per minute. The conversion of volume to number of moles is easy by multiplying the gradient with $[\text{S}_2\text{O}_3^{2-}]$ given in the question stem. Note that volume is given in cm³, hence there is a need to convert this to dm³ for finding the change in number of moles of I_2 per minute.

- 3 (a) (iv) Initial rate = $-\frac{d[\text{I}_2]}{dt} = 7.50 \times 10^{-6} / (10 \times 10^{-3}) = 7.50 \times 10^{-4} \text{ mol dm}^{-3} \text{ min}^{-1}$
 [1] ecf from (a)(iii)

By definition, the **rate of a reaction is always positive**. Since $[\text{I}_2]$ is used up and decreases during the reaction, $d[\text{I}_2]/dt$ is negative. The negative sign is added to the expression to give a positive value for the rate. From (a)(iii), we already found the value of change in number of moles of iodine over time. To get the initial rate required, its simply converting number of moles of iodine to concentration per unit time, by dividing the answer in (a)(iii) with sample volume which is 10cm³. The answer is a positive value.

- 3 (a) (v) Rate = $k[\text{H}^+][\text{propanone}]$
 $k = (7.50 \times 10^{-4}) / (0.2)(0.500) = 7.50 \times 10^{-3} \text{ mol}^{-1} \text{ dm}^3 \text{ min}^{-1}$
 [1] ecf from (a)(iii) for rate and (a)(ii) for rate equation
 [1] correct unit (ecf from rate equation)

Students need to make use of the overall rate equation to find the value of rate constant k . The rate refers to **initial** rate (as we will substitute *initial* $[\text{H}^+]$ and [propanone]), which we have obtained in (a)(iv). By substituting the given initial concentrations of propanone and H^+ for experiment 1, we can obtain the value of k . As rate of reaction is always positive, we do not expect negative value for rate constant k .

- 3 (b) (i) $\text{CH}_3\text{COH}^+\text{CH}_3 \rightarrow \text{H}^+ + \text{CH}_2=\text{C}(\text{OH})\text{CH}_3$ [1]

To write the equation of the rate determining step (r.d.s), students need to recognize that Step 1 + Step 2 + Step 3 will add up and show the overall equation. Students will then need to sieve out the intermediates of this mechanism and ensure that they cancel each other out, by placing them on the correct side of the equation, such that these intermediates will not end up in the final overall equation.

- 3 (b) (ii) Rate = $k[\text{CH}_3\text{COH}^+\text{CH}_3]$
 $K_c = [\text{CH}_3\text{COH}^+\text{CH}_3] / [\text{CH}_3\text{COCH}_3][\text{H}^+]$
 $[\text{CH}_3\text{COH}^+\text{CH}_3] = K_c [\text{CH}_3\text{COCH}_3][\text{H}^+]$
 Rate = $kK_c [\text{CH}_3\text{COCH}_3][\text{H}^+] = k'[\text{CH}_3\text{COCH}_3][\text{H}^+]$
 Therefore, the proposed mechanism is consistent with the rate equation in (a)(ii).

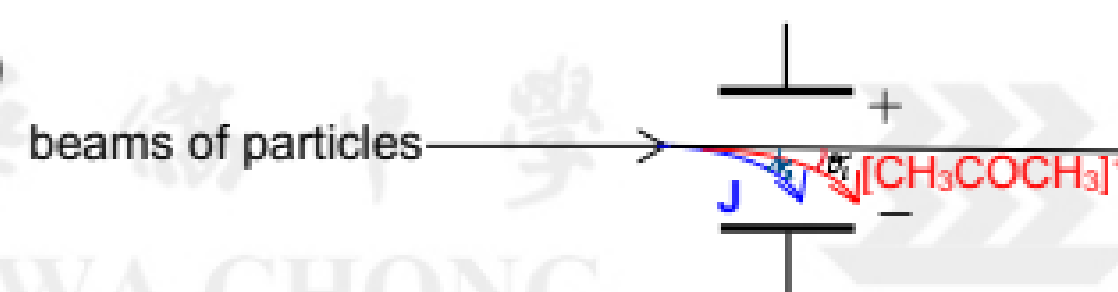
[1]

For a multi-step reaction, the rate equation is derived directly from the stoichiometry of the **slowest step**, i.e. the r.d.s found in (b)(i). From the r.d.s, rate = $k[\text{CH}_3\text{COH}^+\text{CH}_3]$, however $[\text{CH}_3\text{COH}^+\text{CH}_3]$ is an intermediate which does not appear in the overall chemical equation for the reaction, and therefore should **not** appear in the rate equation. We need to substitute $[\text{CH}_3\text{COH}^+\text{CH}_3]$ with the concentration of reactants that produced this intermediate. After substitution, we notice that order of reaction is first with respect to both CH_3COCH_3 and H^+ respectively, thus the proposed mechanism is consistent with the rate equation obtained in (a)(ii).

- 3 (c) (i) The first ionisation energy of propanone is the energy required to remove 1 mole of electrons from 1 mole of $\text{CH}_3\text{COCH}_3(\text{g})$ to form 1 mole of $[\text{CH}_3\text{COCH}_3]^+(\text{g})$. [1]

This question specifically asked for ionisation energy of **propanone**. Students are expected to define based on propanone, rather than giving a generic definition. Do note that propanone is not an atom, but a molecule.

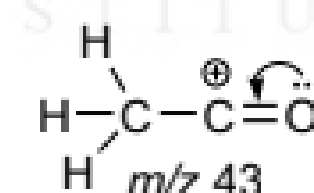
- 3 (c) (ii)



[1] both deflected towards -ve plate
 [1] larger deflection for J

The ionised form of propanone and J are cations, therefore both will be deflected towards the negative plate. As the angle of deflection is directly proportional to the charge (q) to mass (m) ratio, and since q/m for J is larger than that for the ionised form of propanone (J has smaller m), J has a greater angle of deflection with respect to the initial pathway of particles (i.e. $\theta_2 > \theta_1$). Labeling of the ions and sign of plates is crucial. Some students misread the question and gave the path taken by propanone instead of its ionised form.

- 3 (c) (iii)



[1] for arrow & lone pair of electrons

The difference between J and K is that the carbon atom bonded to O has 6 and 8 electrons respectively. This involves the sharing of an electron pair from oxygen, and keeping in mind the fact that a full curly arrow represents the

movement of a pair of electrons, the arrow should begin from the lone pair on O and ended up as a new π bond with C.

- 3 (c) (iv) Mark for idea of one carbon atom exists as ^{13}C OR one hydrogen atom exists as ^2H . [1]

m/z 59 is one unit greater than m/z 58 (given in part (c)). Therefore, that species (with m/z 59) may contain a ^2H atom while that given in (c) contains ^1H , or that species may contain a ^{13}C atom while that given in (c) contains ^{12}C .